Experiment 3(a) - Conduct an experiment to show data visualization using line plot

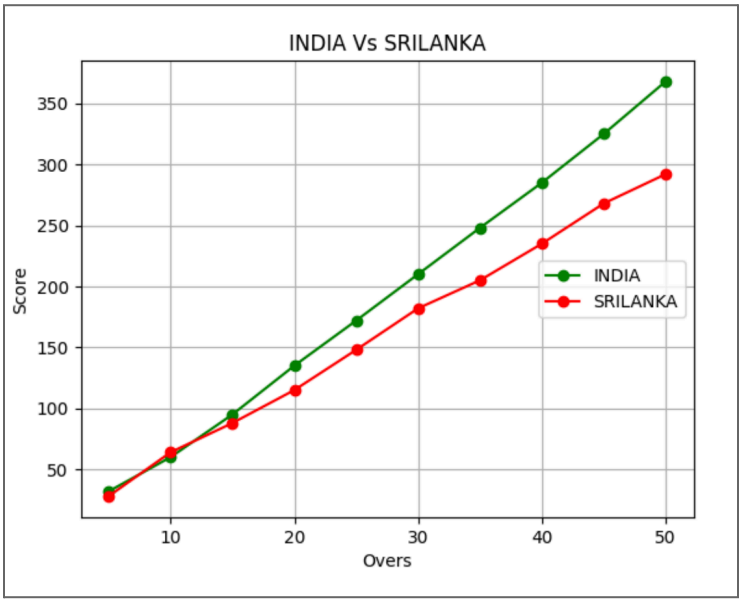
Aim:

To conduct an experiment to show data visualization using a line plot.

Program:

```
import matplotlib.pyplot as cricket
Overs = list(range(5, 51, 5))
Indian_Score = [32, 60, 95, 135, 172, 210, 248, 285, 325, 368]
Srilankan_Score = [28, 64, 88, 115, 148, 182, 205, 235, 268, 292]
cricket.plot(Overs, Indian_Score, color="green", label="INDIA", marker="o")
cricket.plot(Overs, Srilankan_Score, color="red", label="SRILANKA", marker="o")
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend(loc="center right")
cricket.grid(True)
cricket.show()
```

Output:



Result:

Thus, we have conducted an experiment to show data visualization using a line plot.

Experiment 3(b) - Conduct an experiment to show data visualization using bar chart

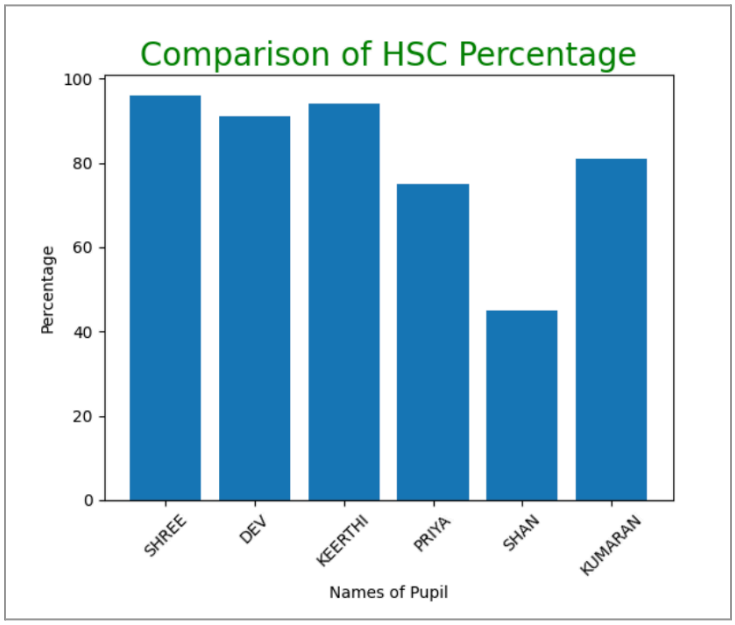
Aim:

To conduct an experiment to show data visualization using a bar chart.

Program:

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

Output:



Result:

Thus, we have conducted an experiment to show data visualization using a bar chart.

Experiment 3(c) - Conduct an experiment to show data visualization using pie chart

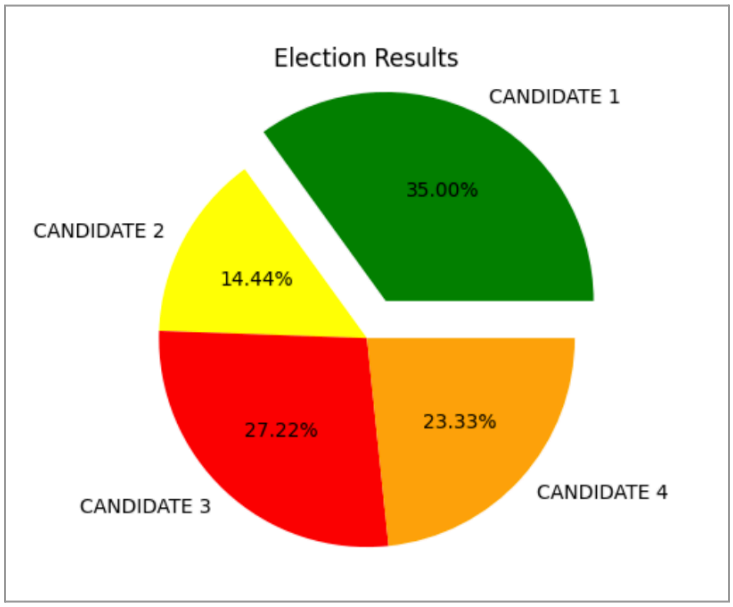
Aim:

To conduct an experiment to show data visualization using a pie chart.

Program:

```
import matplotlib.pyplot as election
labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']
Votes = [315, 130, 245, 210]
colours = ['green', 'yellow', 'red', 'orange']
explode = (0.2, 0, 0, 0)
election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')
election.title('Election Results')
election.show()
```

Output:



Result:

Thus, we have conducted an experiment to show data visualization using a pie chart.